

Rohan Jain

jainrohanj@gmail.com | +1 720-819-0419 | Fremont, CA
linkedin.com/in/rohan-jain11 | github.com/rohanjain11 | rohanjain11.github.io/Rohan-Jain

SUMMARY

MS Data Science graduate from CU Boulder (GPA 3.9/4.0, May 2026) with production experience building backend services and infrastructure for an on-prem LLM agent platform as an AI Engineering Intern at Bright Machines - Kubernetes deployments, security-hardened services, and CI-gated Python packages. Shipped production-grade services across weather forecasting, geospatial routing, and lab infrastructure. Seeking Software Engineer roles focused on backend systems, APIs, and scalable infrastructure.

SKILLS

Languages: Python, TypeScript, R, SQL (PostgreSQL; CTEs, window functions, indexing), Bash; data structures, algorithms, object-oriented design

Backend & APIs: FastAPI, REST API design, OpenAI-compatible APIs, Pydantic v2, SQLAlchemy, Alembic migrations, LangGraph, Model Context Protocol (MCP), Temporal, pytest; error handling, retry/fallback logic

Infrastructure & DevOps: Docker, Docker Compose, Kubernetes, Kustomize, ArgoCD, Azure (AKS, ACR, Entra ID), AWS (Lambda, EC2, RDS, S3, Secrets Manager), oauth2-proxy / Keycloak OIDC, SOPS, GitLab CI/CD, GitHub Actions; Grafana, Loki, Prometheus, Langfuse

Databases & Frontend: PostgreSQL, MySQL, MongoDB, PostGIS, SQLite / sqllite-vec, MinIO, Azure Blob; React, Vite, MUI, Next.js, Streamlit; Git/GitHub

WORK EXPERIENCE

AI Engineering Intern - Bright Machines

Jun 2026 – Present | San Francisco, CA

- Built an on-prem grounded document Q&A platform end to end: **FastAPI** backend, **React/TypeScript** UI, sqllite-vec retrieval, cited answers, and an admin sync worker; **352 automated tests** across 47 commits.
- Designed and shipped a read-only-by-construction **MCP** server from scratch (37 files, +3,163 lines, 12 tools) serving a 32,000-workflow **Temporal** namespace in production; Docker-packaged, deployed via **Kubernetes/Kustomize**.
- Hardened four **MCP** servers behind a production LLM agent: closed a Critical path-traversal vulnerability (**CWE-22**) with 9 regression tests, removed cluster-secret access from a **Kubernetes** service account, enforced least-privilege RBAC.
- Gave the app its first notion of a caller: an **OIDC** identity seam behind oauth2-proxy at the single choke point covering nine admin routes, with both defaults failing closed.
- Raised automated test coverage of a production AI orchestrator from 0 to **135 tests** and wired the suite into GitLab CI, gating every merge request for the first time.

Software Developer - Kopf Lab, University of Colorado Boulder

Dec 2025 – May 2026 | Boulder, CO

- Migrated lab infrastructure to ShinyProxy Operator on AWS EC2 with Keycloak OIDC, per-user container isolation, and Grafana/Loki/Prometheus observability; eliminated a legacy broken nginx sidecar.
- Built an AWS Secrets Manager pipeline to inject OAuth client secrets at deploy time; designed an rsync-based deploy workflow repeatable without GitHub SSH on the server.

Data Science & Engineering Intern - Nexus Weather & Climate

May 2025 – May 2026 | Boulder, CO

- Built an end-to-end reproducible ML pipeline across 6 model types with schema, unit and range validation gates and versioned artifact bundles for auditable experiment comparison.
- Implemented an automated inference quality gate rejecting models whose post-ML MAE fails to beat the pre-ML baseline; reduced failed production runs 90% and saved 20 engineer-hours monthly.

EDUCATION

University of Colorado, Boulder

Aug 2024 – May 2026 | Boulder, CO

Master's in Data Science (GPA: 3.9/4.0)

Mumbai University, Rajiv Gandhi Institute of Technology

Jul 2020 – May 2024 | Mumbai, India

B.E. in Computer Engineering (GPA: 3.7/4.0)

PROJECTS

SafeRide - Geospatial Routing API | Python, FastAPI, PostgreSQL, PostGIS, AWS Lambda

Sep - Dec 2025

- Designed and implemented a REST API serving crash-aware route rankings; built ETL loading crash records and hazard data into PostGIS with composite risk scores via spatial buffer joins, deployed serverlessly on AWS Lambda with RDS.

ClaimPilot AI - Guardrailed LLM Agent | Python, Pydantic v2, OpenAI function calling

Jan 2026

- Architected a hybrid agent for healthcare claims validation: a deterministic Pydantic layer validates codes while OpenAI function calling generates structured fixes; strict schemas with retry logic guarantee a valid report.